



4.6.5 Wrap-Up: Growth Mindset

The power of “YET.”

“The power of believing that you can improve.” - Carol Dweck

Select all of the concepts that apply to your understanding right now.

1. I don’t understand

- isometry
- transformations
- the definition of congruence in terms of rigid motions

YET.

2.

- Isometry
- Mapping
- One-to-one correspondence
- Definition of congruence in terms of rigid motions

doesn’t

make sense to me YET.

3. I can explain

- isometry.
- mapping.
- one-to-one correspondence.
- transformations.
- the definition of congruence in terms of rigid motions.

Unit 4, Lesson 7: Justifying Congruence



Benchmarks

MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself, or onto another congruent or similar figure.

MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.



Learning Targets

- ☐ I can describe a sequence of transformations that will map a preimage and an image to show congruence.
- ☐ I can describe a sequence of transformations that map one figure onto another congruent figure.



4.7.1 Warm-Up: You Can Do It!

"You should be far more concerned with your current trajectory than with your current results." - James Clear

Has anyone ever asked you, "What goals do YOU want to achieve in your lifetime?"



Setting goals is linked to self-confidence, motivation, and autonomy. A 2015 study showed people who wrote down their goals were 33 percent more successful in achieving them than those who just thought about the outcomes. Making goals can be fun.

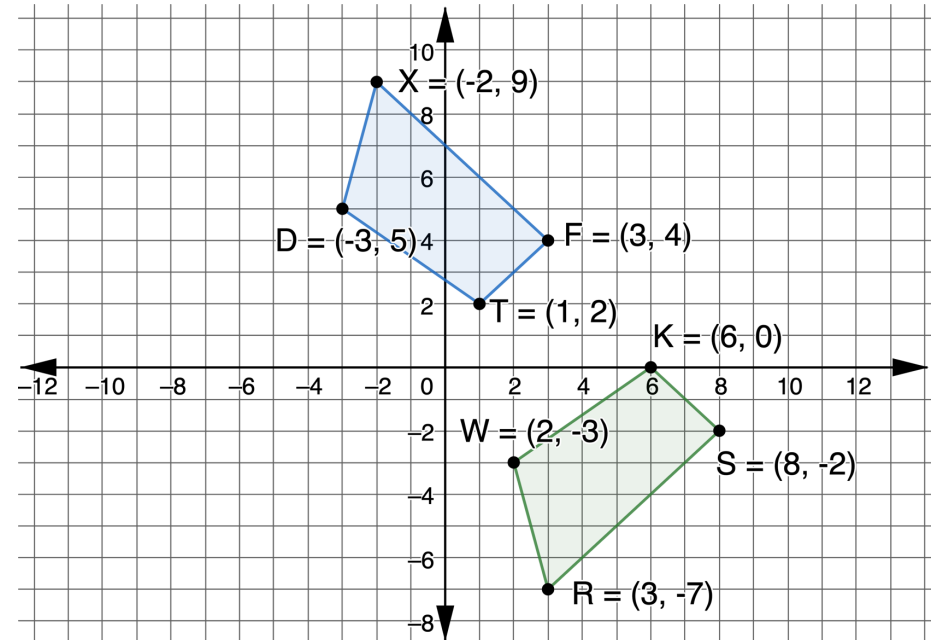
For example, you can name places you would like to visit, activities you would like to do, or people you would like to meet. By setting the goal, you have already set your trajectory toward achieving it. In the table, write three life goals and two goals for the rest of this school year.

Life Goals	Goals for this school year
"If you are always trying to be normal, you will never know how amazing you can be." Maya Angelou	



4.7.2 Guided Instruction: Justifying Congruence

1. Quadrilaterals $XFTD$ and $KSRW$ are shown on the coordinate plane.



- a. Describe a sequence of transformations that will map quadrilateral $KSRW$ onto quadrilateral $XFTD$.

- b. Complete the table that shows which vertices were mapped to each other when quadrilateral $KSRW$ was mapped onto quadrilateral $XFTD$.

Quadrilateral $KSRW$	Quadrilateral $XFTD$
K	
S	
R	
W	



Definition of Congruence in Terms of Rigid Motions

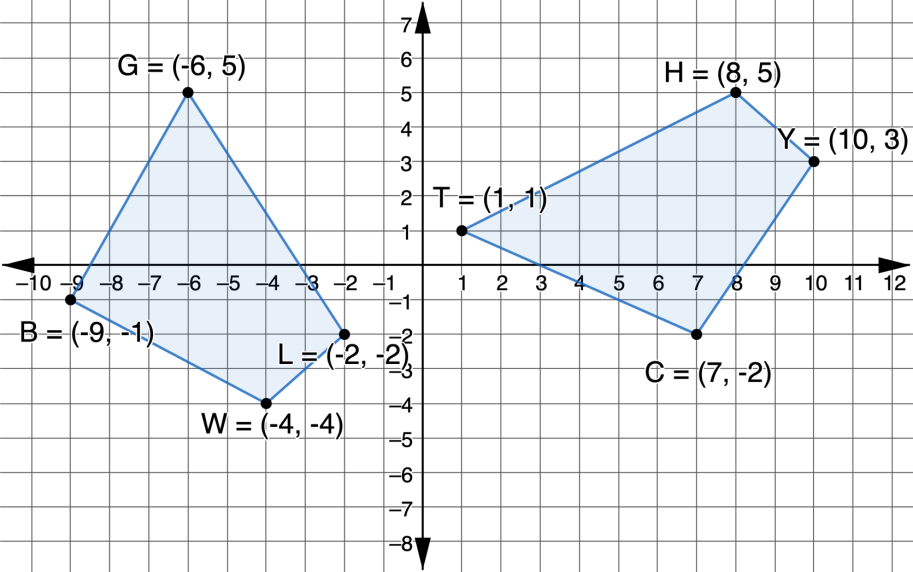
Two figures are congruent if and only if there exists one or more rigid motions, which will map one figure onto the other.

- c. Complete the statements.

Quadrilateral $KSRW$ is congruent to quadrilateral _____ because the sequence of transformations that can be used to map the quadrilaterals were _____.

Because vertex K maps to vertex _____, vertex S maps to vertex _____, vertex R maps to vertex _____, and vertex W maps to vertex _____. The definition of congruence in terms of rigid motions justifies that $KSRW \cong$ _____.

2. Quadrilaterals $BGLW$ and $HYCT$ are shown on the coordinate plane.



- a. Use algebraic descriptions to describe a sequence of transformations that will map quadrilateral $BGLW$ onto quadrilateral $HYCT$.
- b. Complete the table that shows which vertices were mapped to each other when quadrilateral $BGLW$ was mapped onto quadrilateral $HYCT$.

Quadrilateral $BGLW$	Quadrilateral $HYCT$
B	
G	
L	
W	



When using rigid motions to map one figure onto the other, always be sure the corresponding vertices match before stating congruence.

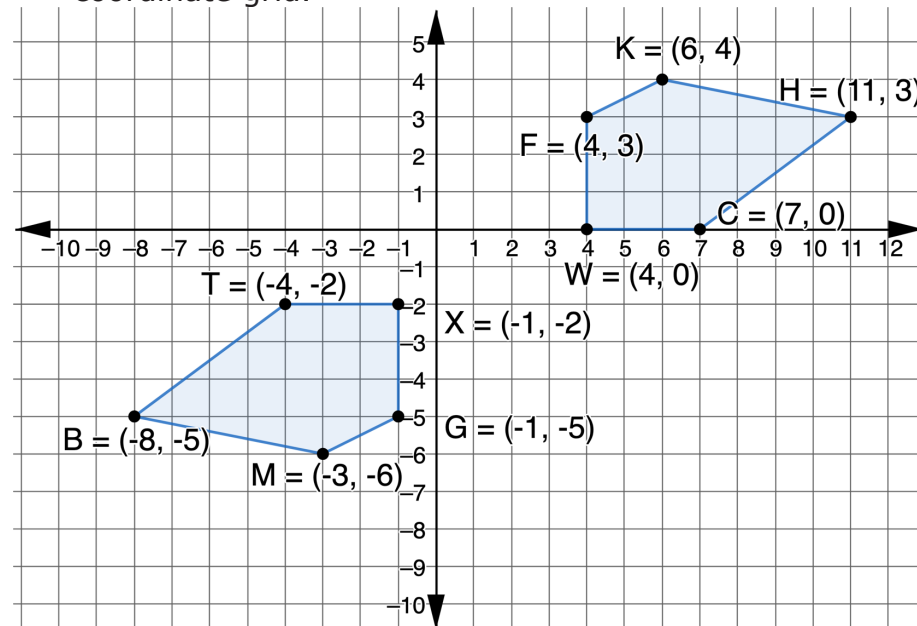
- c. Complete the congruence statement.

$BGLW \cong$ _____



4.7.3 Guided Practice: Justifying Congruence

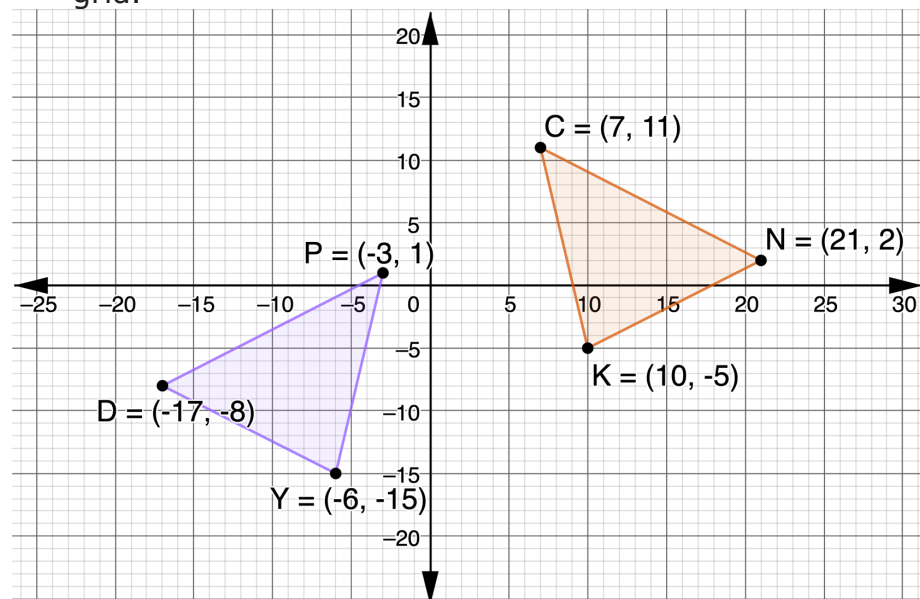
1. Polygons $TXGMB$ and $WFKHC$ are shown on the coordinate grid.



- a. Use algebraic descriptions to describe a sequence of transformations that will map polygon $TXGMB$ onto polygon $WFKHC$.
- b. Complete the congruence statement.

$TXGMB \cong$ _____

2. Triangles CKN and PYD are shown on the coordinate grid.



- a. The sequence of transformations that will map

$\triangle CKN$ onto $\triangle PYD$ is

- translating $\triangle CKN$ left 14,
- reflecting $\triangle CKN$ across the y -axis,
- rotating $\triangle CKN$ 270° clockwise about the origin,

then

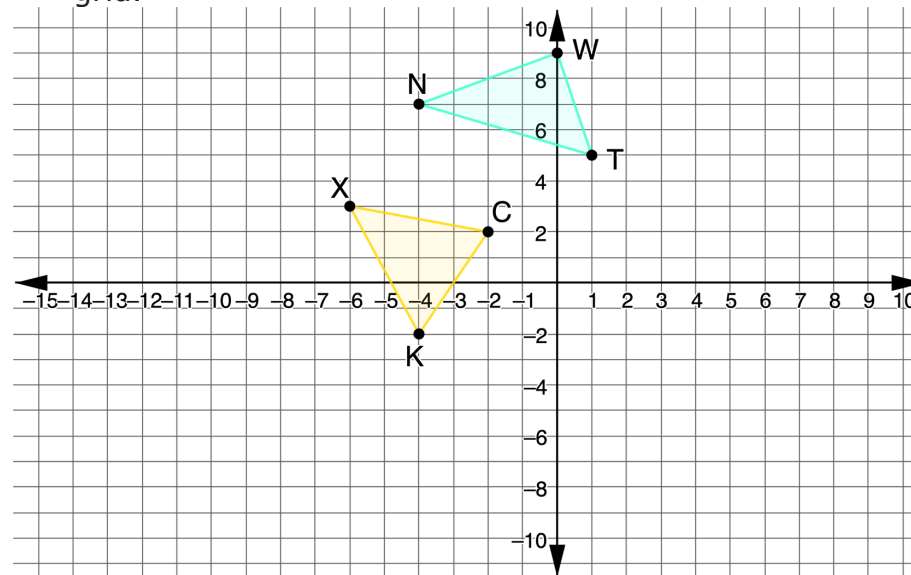
- translating $\triangle C'K'N'$ right 4, down 10.
- translating $\triangle C'K'N'$ right 10, down 4.
- reflecting $\triangle C'K'N'$ across the x -axis.

- b. Vertex C maps to vertex _____, vertex K maps to vertex _____, and vertex N maps to vertex _____ therefore $\triangle CKN$ is congruent to $\triangle$ _____ by the definition of congruence in terms of rigid motions.



4.7.4 Your Turn!

1. Triangles XCK and TWN are shown on the coordinate grid.



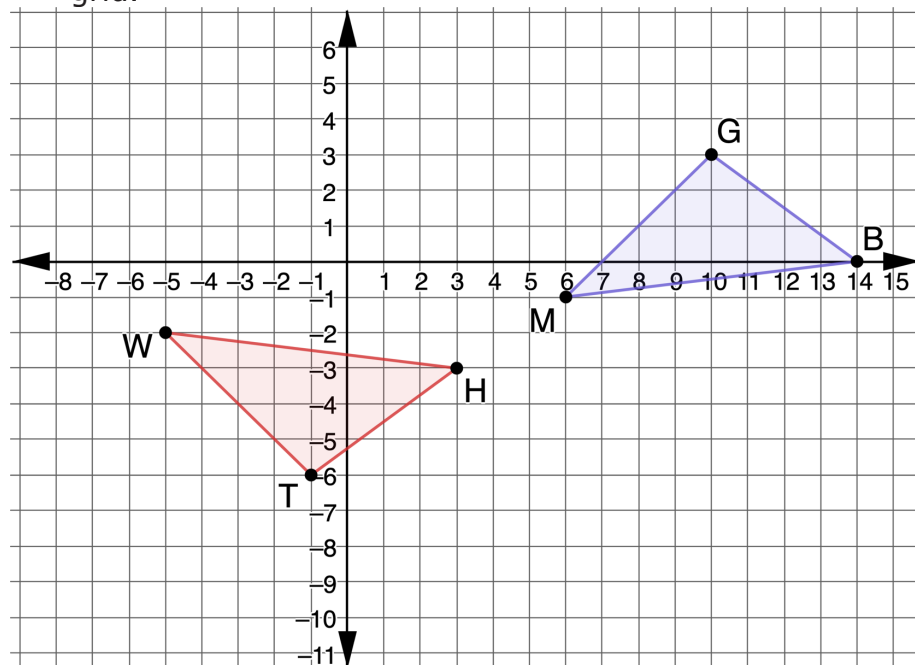
Use algebraic descriptions to describe a sequence of three transformations that will map $\triangle XCK$ onto $\triangle TWN$.

Transformations
$(x, y) \rightarrow$ _____
$(x, y) \rightarrow$ _____
$(x, y) \rightarrow$ _____



4.7.5 Wrap-Up: Ticket Out the Door

1. Triangles TWH and GBM are shown on the coordinate grid.



- a. Use written descriptions or algebraic descriptions to describe a sequence of two transformations that will map the two triangles onto each other.
- b. Complete the table that shows which vertices were mapped to each other.

$\triangle TWH$	$\triangle GBM$
T	
W	
H	

- c. Complete the congruence statement.

$$\triangle TWH \cong \underline{\hspace{2cm}}$$



Unit 4, Lesson 8: Putting It All Together



Benchmarks

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.

MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.



Learning Targets

- ☐ I can determine a sequence of transformations that map one figure onto another congruent figure.
- ☐ I can match the written and algebraic descriptions of a sequence of transformations given the pre-image and image of a figure.